

PROGRAM

SENDER

```
#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <stdio.h>

int main() {
    key_t key = ftok("shmfile", 65);

    // Create shared memory
    int shmid = shmget(key, 1024, 0666|IPC_CREAT);

    // Attach
    char *str = (char*) shmat(shmid, (void*)0, 0);

    printf("Write Data : ");
    fgets(str, 100, stdin); // Write to shared memory

    printf("Data written in memory: %s\n", str);

    // Detach
    shmdt(str);
    return 0;
}
```

RECEIVER

```
#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>

int main() {
    key_t key = ftok("shmfile", 65);

    // Open shared memory
    int shmid = shmget(key, 1024, 0666|IPC_CREAT);

    // Attach
    char *str = (char*) shmat(shmid, (void*)0, 0);

    printf("Data read from memory: %s\n", str);

    // Detach
    shmdt(str);

    // Remove shared memory
    shmctl(shmid, IPC_RMID, NULL);
    return 0;
}
// Note: Create a dummy file named 'shmfile' before running.
```

OUTPUT

```
blackie@blackie-pc:~/GEC_PKD_IT/S4/OS_LAB/EXPERIMENT_4$ gcc 4d_sender.c
```

```
blackie@blackie-pc:~/GEC_PKD_IT/S4/OS_LAB/EXPERIMENT_4$ ./a.out
```

```
Write Data : HELLO
```

```
Data written in memory: HELLO
```

```
blackie@blackie-pc:~/GEC_PKD_IT/S4/OS_LAB/EXPERIMENT_4$ gcc 4d_receiver.c
```

```
blackie@blackie-pc:~/GEC_PKD_IT/S4/OS_LAB/EXPERIMENT_4$ ./a.out
```

```
Data read from memory: HELLO
```

```
blackie@blackie-pc:~/GEC_PKD_IT/S4/OS_LAB/EXPERIMENT_4$
```